

Algebra 1

Unit 1

Lesson 5 Homework

Name **Answer Key** _____

Date _____

Period _____

1. In radical form, the expression $x^{\frac{1}{2}}$ is equivalent to $\sqrt{x}$.

2. In exponential form, the expression $\sqrt[3]{x}$ is equivalent to $x^{\frac{1}{3}}$.

For questions 3-5, rewrite each exponential expression as a radical expression. Then, evaluate the expression.

3. $(-125)^{\frac{1}{3}} = \sqrt[3]{-125} = -5$

4. $(1024)^{\frac{1}{5}} = \sqrt[5]{1024} = 4$

5. $49^{\frac{1}{2}} = \sqrt{49} = 7$

6. Simplify the expression using the laws of exponents and then find the value.

$$36^{-\frac{5}{4}} \cdot 36^{\frac{7}{4}} = 36^{\frac{1}{2}} = \sqrt{36} = 6$$

The expression $\frac{81^{\frac{10}{12}}}{81^{\frac{7}{12}}}$ can be rewritten using the law of exponents.

7. Which law of exponents can be used to rewrite the expression?
Quotient of powers

8. Rewrite the expression as an equivalent expression.

$$81^{\frac{3}{12}} = 81^{\frac{1}{4}}$$

9. Write simplified expression in radical form.

$$\sqrt[4]{81}$$

10. Evaluate the radical expression.

$$\sqrt[4]{81} = 3$$